

POSITION AND FORCE ANALYSIS OF A 3-DOF PARALLEL
TRANSLATIONAL MECHANISM

By

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To my mother, Jinyu Ma and my father, Wuxu Zhao

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LIST OF ABBREVIATIONS

DOF	Degrees of freedom
PM	Parallel mechanism
3D	Three-dimensional
EE	End-effector
P	Prismatic joint
R	Revolute joint
3- <u>P</u> RRR	A mechanism has three identical legs, each one having an actuated prismatic joint and three revolute joints.

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In the past several decades, parallel translational mechanisms have been studied in many areas. Many novel designs have been proposed and they commonly provide high stiffness, accuracy, and high speed and acceleration. This thesis mainly focuses on the position and force analysis of the Tripteron, a three degree of freedom parallel translational mechanism. According to the special architecture, this robot can be designed as several versions and provide a simple and intuitive input-output motion.

In this thesis, due to the numerous variations of the mechanical architecture, a selected 3D model is designed and presented. Then in order to solve the position problem, the closed-loop vector equation and the general kinematic analysis method are used. Furthermore, the notation of the wrench and the free body diagram of the end-effector are utilized to solve the reverse force analysis problem. Finally, all details of the design are shown by CAD drawings.

CHAPTER 1 INTRODUCTION

With the development of robotic systems, parallel mechanisms (PMs) have been studied in many areas such as new robot design, position analysis, and kinematic analysis. The main reason for this enthusiasm is that PMs often provide higher accuracy, velocity, and acceleration [1]. Furthermore, a successful design of the parallel manipulator will provide numerous commercial opportunities. The analysis of PMs, however, commonly has more difficulties due to the complex mechanical architecture.

In the family of translational parallel robots, the most famous and successful mechanism is the Delta robot. Several productions of the robot have been used in industries like the packaging and the surgical areas [2]. On the other hand, the special design of the Delta robot has attracted a lot of interest in academia. Many modified versions and analysis methods [3] have been published by universities and institutes. However, the complex motion of the whole mechanism leads to complicated analysis approaches.

According to this, a new 3-DOF parallel translational mechanism, called the Tripteron, was first designed by researchers at Laval University [4]. This novel robot not only exhibits high accuracy, speed and acceleration, but changes the complicated motion to a simple one whereby three orthogonal moving directions are driven by three independent linear actuators. Therefore, the analysis of the Tripteron will be intuitive but also challenging.

The rest of this thesis is developed as follows. Chapter 2 presents the literature review of this study. It presents some papers and researches about the general kinematic analysis of mechanisms, the Delta robot, and the Tripteron. Then in chapter

3, the architecture and the mobility of the Tripteron are presented. Also, the selected design of the Tripteron is shown. Furthermore, position and force analysis are discussed in the form of the forward and reverse static analysis in chapter 4. Finally, the details of the model design drawing are shown in appendix.

CHAPTER 2 LITERATURE REVIEW

Overview

The literature review discusses work related to this thesis in the following areas: Kinematic Analysis and Screw Theory, History of the Delta Robot, and Development of the Tripteron.

Kinematic Analysis and Screw Theory

Crane and Duffy (1998) authored a book on the kinematic position analysis of robot manipulators. The book introduced notation for links that are interconnected serially by joints. It presented equations for the degrees of freedom (DOF) of manipulators, and forward and reverse position analysis for serial robot arms. The material was also useful for modeling parallel mechanisms. This text also discussed spherical closed-loop mechanisms and quaternions.

Rao (2006) wrote an excellent text on kinetics of particles and rigid bodies. In this book, Rao uniquely blended concepts, theories and examples at an easy learning level. This work was able to develop a systematic approach to solve kinematics and dynamics problem.

Crane, Griffis, and Duffy (2019) discussed screw theory and its application to spatial robot manipulators. The authors presented reciprocal screws, and the concept of the dyname and wrench which was very useful for analyzing forces and moments of a rigid body. This book also analyzed the velocity, acceleration and singularity of serial manipulators which were useful for studying the reverse analysis of the Tripteron.

History of the Delta Robot

Bonev (2001) introduced the history of the famous 3-DOF parallel mechanism, the Delta robot. The author showed that the Delta robot is successful in many areas such as the packaging industry and the surgical domain.

Li and Xu (2005) designed a modified Delta robot and analyzed the dynamics of it for cardiopulmonary resuscitation. In the inverse dynamic analysis part, the principle of virtual work and the computed torque method were implemented.

Development of the Tripteron

Gosselin, Kong, Foucault and Bonev (2004) first presented a 3 degree of freedom translational parallel mechanism, called the Tripteron. The authors theoretically analyzed the direct kinematics and singularities of the Tripteron and built a prototype.

Gosselin, Masouleh, Duchaine, Richard, Foucault and Kong (2007) provided the kinematic architectures and benchmarking of parallel mechanisms of the multipteron family. They built 2 prototypes to test the benchmarking and experimental characterization of the Tripteron and Quadrupteron. The conclusion that the Tripteron did not have any constraint singularities was very important to the design and testing of the 3D model.

Gosselin (2009) proposed the compact dynamic models for the Tripteron and Quadrupteron. Though this paper focused on the dynamic models of the two parallel manipulators, the presentation of the architecture and kinematics of these robots provided excellent examples of the 3D model of the Tripteron and Quadrupteron for further study.

Yahyapour, Hasanvand, Masouleh, Yazdani and Tavakoli (2013) established a new approach for the dynamic model of the Tripteron. The paper proposed an algorithm

by separating the Tripteron into four subsystems and discussed results of some trajectory cases using the MD ADAMS-View software.

Seward and Bonev (2014) presented a new 6-DOF parallel robot with direct kinematics similar to the Tripteron. The authors established a 6-PRRR translational parallel manipulators to provide 6-DOF. This robot was extremely valuable in future research for designing more types of parallel mechanisms.

Anvari, Ataei and Tale-Masouleh (2018) proposed a new geometric method to test intersection segment by segment for avoiding collision of any parts of robot. In the text, though this method was for all serial robots. However, it was implemented on a Tripteron robot and gave some types of collisions of the mechanism. This information was valuable for designing and testing the Tripteron device.

Arian, Dnanei and Tale-Masouleh (2018) analyzed the kinematics and dynamics of the Tripteron using the Newton-Euler approach. The authors modified the joints of the end-effector (EE) of the mechanism in order to solve the dynamical constraints and then derived the forces of the actuator in a dynamic analysis section. Especially important was the use of the geometric relationship of the upper and lower links of the mechanism.

CHAPTER 3 DESIGN OF THE TRIPTERON MECHANISM

Overview

As shown in Figure 3-1, the Tripteron is a translational parallel mechanism (PM) comprised of three kinematically identical legs [5,6], each one having a lower link, an upper link, a linearly actuated prismatic joint (P) fixed at an orthogonal coordinate frame, and three revolute joints (R) whose axes are parallel to each other. Then the three ending revolute joints are connected to an end-effector (EE), but their axes are orthogonal to each other.

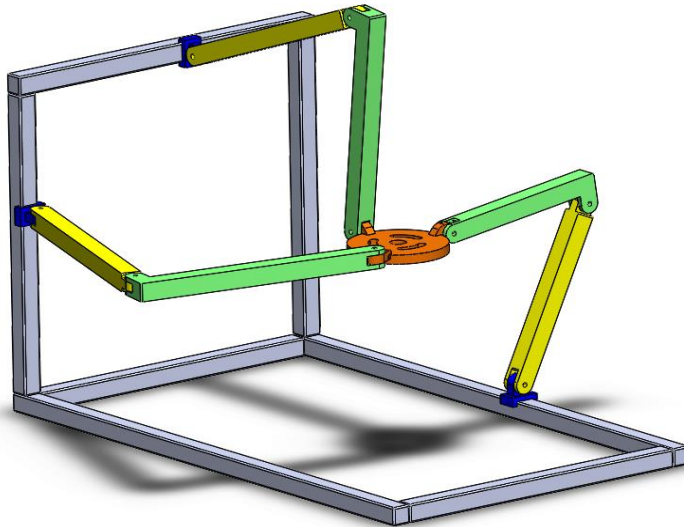


Figure 3-1. A simple model of the Tripteron.

The Tripteron is a fully decoupled mechanism [4,7] which means that each linear actuator on the prismatic joint will provide only one translational degree of freedom. Moreover, each of the total three degrees of freedom is parallel to the direction of movement of the corresponding actuator but perpendicular to the others. Thus, the robot can be described as a 3-DOF 3-PRRR translational PM. However, the theoretical mobility of the Tripteron is odd and can be calculated by Equation 3-1 [9].

$$M = 6(n - 1) - \sum_{i=1}^j (6 - f_i) = 6(11 - 1) - \sum_{i=1}^{12} (6 - 1) = 0 \quad (3-1)$$

where M is the total mobility, n is the number of bodies of the system, j represents the number of joints in the mechanism, and f_i denotes the number of relative degrees of freedom permitted by joint i . In this case, the theoretical mobility of the mechanism is zero. Therefore, by this equation the Tripteron is an over-constrained structure [1,8]. However, due to its special geometry, it can be shown that the device has three degrees of freedom. Furthermore, as proven by previous research, it can be concluded that the mechanism has no singularities [1,6] inside its workspace.

Selected Design

Since the axes of three prismatic joints just need to be orthogonal to each other and a cuboid frame has 12 edges, there can be 64 variations of the Tripteron, i.e. there are 64 possible configurations of the links that will position the end-effector at the same location. Some of them are shown in Figure 3-2.

Limited by the shape and size of the workspace, which for this case is a $40 \times 40 \times 30$ cuboid¹ on a platform, a 3D model of the selected design is shown in Figure 3-3. Considering the size of the platform and the kinematics of the mechanism, some parameters of the designed 3D model should be specially selected such as the frame, each link of the three legs, and the end-effector. All the details of the model design drawing are shown in appendix.

¹ The units of length are arbitrary as long as the selected unit of length is used consistently in the analysis.

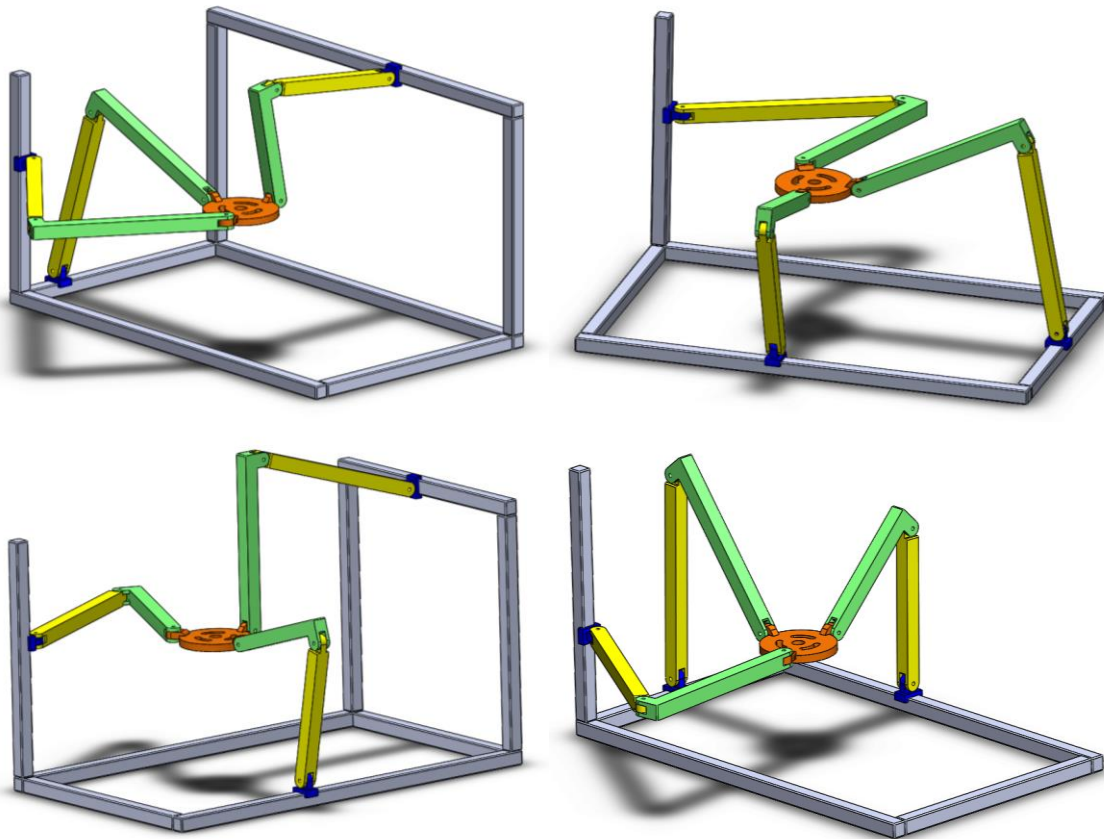


Figure 3-2. Some variations of the Tripteron.

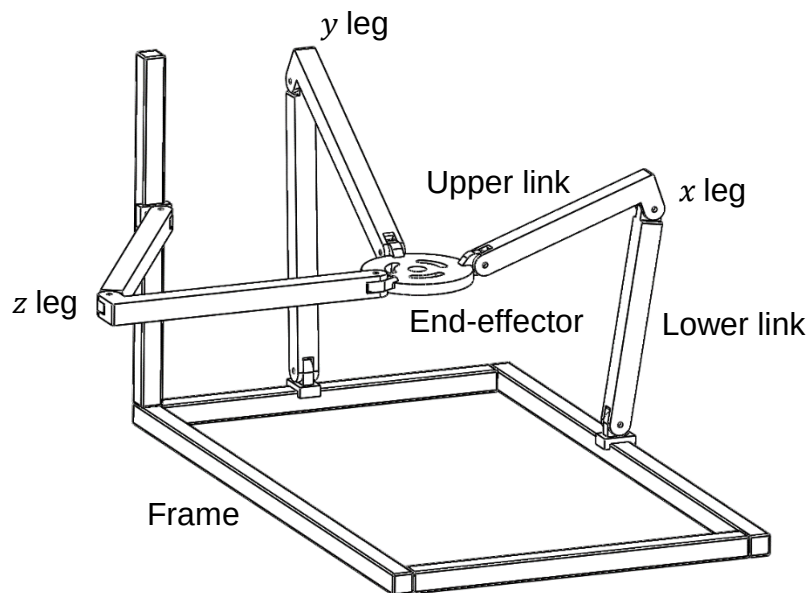


Figure 3-3. A 3D model of the selected design of the Tripteron.

CHAPTER 4 POSITION AND FORCE ANALYSIS

Overview

Although the Tripteron has a very simple kinematical mechanism, it is extremely valuable for building and testing a prototype to find out the position of the EE and the force that actuators should apply to maintain static equilibrium. Therefore, this chapter contains the forward and reverse position analysis of the EE and the reverse force analysis which determines the force distribution on the three prismatic joints when a force and torque (wrench) is applied to the EE.

Position Analysis

Forward Analysis

As shown in Figure 4-1, given the origin of the coordinate of whole mechanism and the coordinates of the three prismatic joints, then the vector of the EE can be easily derived as $\mathbf{r} = (x_{EE}, y_{EE}, z_{EE})^T = (x_1, y_2, z_3)^T$ [5,7,10].

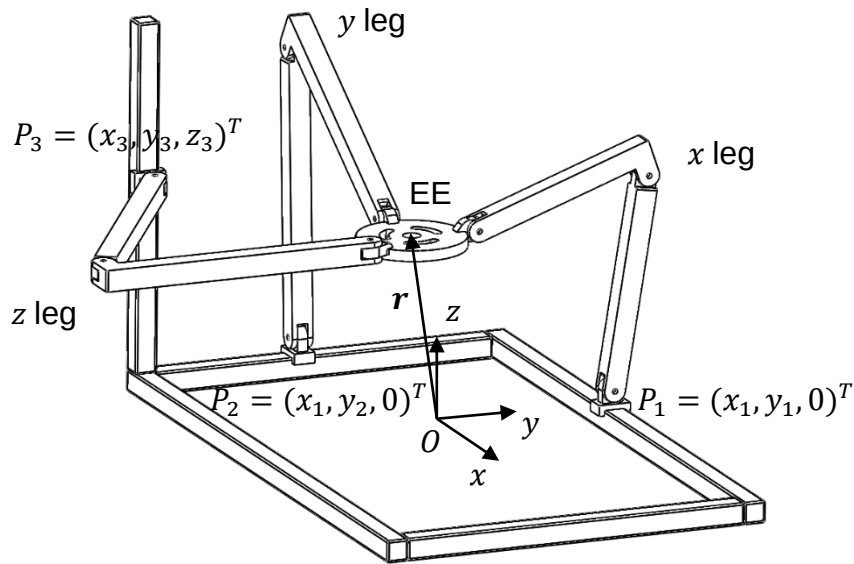


Figure 4-1. Positions of the three prismatic joints of the Tripteron.

Then, the velocity and acceleration of the EE are simply equal to the first and second time derivative of position vectors.

Reverse Analysis

The forward position analysis, though, is simple and direct. It is mathematically and geometrically complex to get the position, velocity and acceleration of the three prismatic joints when one only knows the position of the EE.

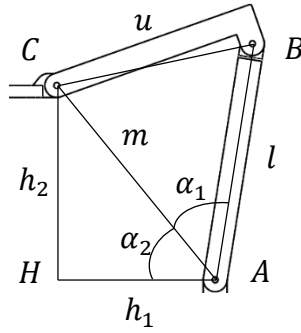


Figure 4-2. Geometric relationship of a leg of the Tripteron.

Before solving the reverse problem, the solution of the following geometric problem of the three identical legs should be completed. As shown in Figure 4-2, vertices A , B and C , which form triangle ΔABC with edges l , u , and m , are the center of connection of links and H is the perpendicular foot of the right triangle ΔAHC with edges h_1 and h_2 . Then the two unknown angles α_1 and α_2 can be derived from the geometric relationship and the law of cosines [1] as follows:

$$\alpha_1 = \cos^{-1} \left(\frac{m^2 + l^2 - u^2}{2ml} \right) \quad (4-1)$$

$$\alpha_2 = \tan^{-1} \left(\frac{h_2}{h_1} \right) \quad (4-2)$$

$$\theta = \alpha_1 + \alpha_2 \quad (4-3)$$

Note that as shown in Equations 4-1 and 4-2, angles α_1 and α_2 will both have two values because of the inverse trigonometric functions. However, due to the geometric constraints only values between 0 and 90 degrees are realizable angles.

Reverse displacement analysis

As shown in Figure 4-3, the position vector of the EE is given as $\mathbf{r} = (x_{EE}, y_{EE}, z_{EE})^T$ and other vectors represent connections of the mechanism. The unit vectors of x , y , and z directions are $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$, respectively.

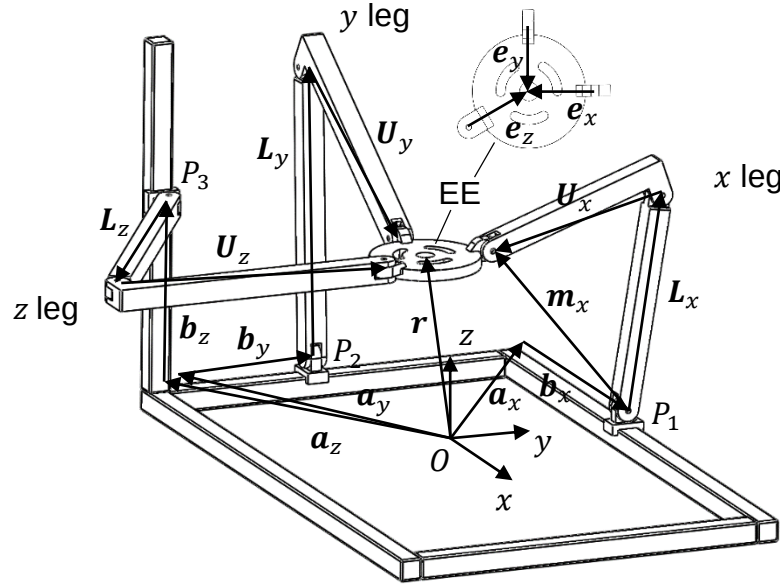


Figure 4-3. Vectors of all links and connections of the Tripteron.

According to the fact that all the $\mathbf{a}_i$ vectors are fixed, the positions of the three prismatic joints can be easily obtained by the given vector $\mathbf{r}$. Therefore, the coordinate of the three prismatic joints can be written respectively as:

$$P_1 = (\mathbf{r} \cdot \mathbf{e}_1, \mathbf{a}_x \cdot \mathbf{e}_2, \mathbf{a}_x \cdot \mathbf{e}_3)^T = (x_{EE}, \mathbf{a}_x \cdot \mathbf{e}_2, 0)^T \quad (4-4)$$

$$P_2 = (\mathbf{a}_y \cdot \mathbf{e}_1, \mathbf{r} \cdot \mathbf{e}_2, \mathbf{a}_y \cdot \mathbf{e}_3)^T = (\mathbf{a}_y \cdot \mathbf{e}_1, y_{EE}, 0)^T \quad (4-5)$$

$$P_3 = (\mathbf{a}_z \cdot \mathbf{e}_1, \mathbf{a}_z \cdot \mathbf{e}_2, \mathbf{r} \cdot \mathbf{e}_3)^T = (\mathbf{a}_z \cdot \mathbf{e}_1, \mathbf{a}_z \cdot \mathbf{e}_2, z_{EE})^T \quad (4-6)$$

Moreover, since one of the three parts forms a closed-loop geometry, positions of other joints can be denoted with vectors as in the following steps. Taking the x leg as an example, the closed-loop vector equation [9] can be expressed as:

$$\mathbf{a}_x + \mathbf{b}_x + \mathbf{L}_x + \mathbf{U}_x + \mathbf{e}_x - \mathbf{r} = \mathbf{0} \quad (4-7)$$

where $\mathbf{a}_x$ is fixed and known and $\mathbf{r} = (x_{EE}, y_{EE}, z_{EE})^T = x_{EE}\mathbf{e}_1 + y_{EE}\mathbf{e}_2 + z_{EE}\mathbf{e}_3$. As the geometric relationships shown, the direction of unknown vector $\mathbf{b}_x$ is only along the x direction. Therefore, the norm of $\mathbf{b}_x$ can be derived from equation 4-7 by taking dot products with $\mathbf{e}_1$ on both sides for eliminating other unknown terms:

$$\|\mathbf{b}_x\| = x_{EE} - \mathbf{a}_x \cdot \mathbf{e}_1 \quad (4-8)$$

$$\mathbf{b}_x = \|\mathbf{b}_x\|\mathbf{e}_1 = (x_{EE} - \mathbf{a}_x \cdot \mathbf{e}_1)\mathbf{e}_1 \quad (4-9)$$

Then, considering that vector $\mathbf{m}_x = \mathbf{L}_x + \mathbf{U}_x$ and putting Equation 4-9 into Equation 4-7 will lead to:

$$\mathbf{m}_x = \mathbf{r} - \mathbf{a}_x - \mathbf{b}_x - \mathbf{e}_x = y_{EE}\mathbf{e}_2 + z_{EE}\mathbf{e}_3 - \mathbf{a}_x - \mathbf{e}_x + (\mathbf{a}_x \cdot \mathbf{e}_1)\mathbf{e}_1 \quad (4-10)$$

where all vectors in the rightmost hand side are known. And two edges h_{1x} and h_{2x} can be obtained as:

$$h_{1x} = -y_{EE} + (\mathbf{a}_x + \mathbf{e}_x) \cdot \mathbf{e}_2 \quad (4-11)$$

$$h_{2x} = z_{EE} \quad (4-12)$$

Since the lengths of links are known, using the results of geometric problem above will lead to:

$$\alpha_{1x} = \cos^{-1} \left(\frac{\|\mathbf{m}_x\|^2 + \|\mathbf{L}_x\|^2 - \|\mathbf{U}_x\|^2}{2\|\mathbf{m}_x\|\|\mathbf{L}_x\|} \right) \quad (4-13)$$

$$\alpha_{2x} = \tan^{-1} \left(\frac{h_{2x}}{h_{1x}} \right) \quad (4-14)$$

$$\theta_x = \alpha_{1x} + \alpha_{2x} \quad (4-15)$$

And a similar method can be used for the y and z legs. Then, vectors of the lower and upper links can be expressed as:

$$\mathbf{L}_x = \|\mathbf{L}_x\|(-\cos\theta_x\mathbf{e}_2 + \sin\theta_x\mathbf{e}_3) \quad (4-16)$$

$$\mathbf{U}_x = \mathbf{m}_x - \mathbf{L}_x \quad (4-17)$$

Similarly, considering that the three legs are identical, the reverse analysis of the y and z legs is easily obtained. For the y leg, equations of $\mathbf{b}_y$, $\mathbf{m}_y$, h_{1y} , h_{2y} , $\mathbf{L}_y$, and $\mathbf{U}_y$ can be written respectively as:

$$\mathbf{b}_y = (y_{EE} - \mathbf{a}_y \cdot \mathbf{e}_2)\mathbf{e}_2 \quad (4-18)$$

$$\mathbf{m}_y = x_{EE}\mathbf{e}_1 + z_{EE}\mathbf{e}_3 - \mathbf{a}_y - \mathbf{e}_y + (\mathbf{a}_y \cdot \mathbf{e}_2)\mathbf{e}_2 \quad (4-19)$$

$$h_{1y} = x_{EE} + (-\mathbf{a}_y + \mathbf{e}_y) \cdot \mathbf{e}_1 \quad (4-20)$$

$$h_{2y} = z_{EE} \quad (4-21)$$

$$\mathbf{L}_y = \|\mathbf{L}_y\|(\cos\theta_y\mathbf{e}_1 + \sin\theta_y\mathbf{e}_3) \quad (4-22)$$

$$\mathbf{U}_y = \mathbf{m}_y - \mathbf{L}_y \quad (4-23)$$

It is the same for z leg:

$$\mathbf{b}_z = (\mathbf{r} \cdot \mathbf{e}_3)\mathbf{e}_3 = z_{EE}\mathbf{e}_3 \quad (4-24)$$

$$\mathbf{m}_z = x_{EE}\mathbf{e}_1 + y_{EE}\mathbf{e}_2 - \mathbf{a}_z - \mathbf{e}_z \quad (4-25)$$

$$h_{1z} = y_{EE} - (\mathbf{a}_z + \mathbf{e}_z) \cdot \mathbf{e}_2 \quad (4-26)$$

$$h_{2z} = x_{EE} - (\mathbf{a}_z + \mathbf{e}_z) \cdot \mathbf{e}_1 \quad (4-27)$$

$$\mathbf{L}_z = \|\mathbf{L}_z\|(\cos\theta_z\mathbf{e}_2 + \sin\theta_z\mathbf{e}_1) \quad (4-28)$$

$$\mathbf{U}_z = \mathbf{m}_z - \mathbf{L}_z \quad (4-29)$$

Velocity analysis

According to the translational parallel movement of the EE, the vector of the velocity $\mathbf{v}_{EE}$ is defined by the time derivative of the position vector $\mathbf{r}$ as shown in Equation 4-30.

$$\mathbf{v}_{EE} = (\dot{x}_{EE}, \dot{y}_{EE}, \dot{z}_{EE})^T = \dot{x}_{EE}\mathbf{e}_1 + \dot{y}_{EE}\mathbf{e}_2 + \dot{z}_{EE}\mathbf{e}_3 \quad (4-30)$$

Moreover, since the lower or upper link of each leg will only rotate relative to its joint, the velocity of each link is denoted as the cross product of the angular velocity and the position vector. Taking the x leg as an example, velocities can be shown as [11,12]:

$$\dot{\mathbf{L}}_x = \boldsymbol{\omega}_{L_x} \times \mathbf{L}_x \quad (4-31)$$

$$\dot{\mathbf{U}}_x = \boldsymbol{\omega}_{U_x} \times \mathbf{U}_x \quad (4-32)$$

where $\boldsymbol{\omega}_{L_x}$ and $\boldsymbol{\omega}_{U_x}$ are the vectors of the angular velocity of the lower and upper links of x leg, respectively. Taking the time derivative of Equation 4-7 and inserting Equations 4-30, 4-31 and 4-32 into it will lead to:

$$\dot{\mathbf{b}}_x + \boldsymbol{\omega}_{L_x} \times \mathbf{L}_x + \boldsymbol{\omega}_{U_x} \times \mathbf{U}_x - (\dot{x}_{EE}\mathbf{e}_1 + \dot{y}_{EE}\mathbf{e}_2 + \dot{z}_{EE}\mathbf{e}_3) = \mathbf{0} \quad (4-33)$$

Taking the dot product of $\mathbf{e}_1$, $\mathbf{U}_x$, and $\mathbf{L}_x$ on both sides of Equation 4-33 and using the property of the scalar triple product will yield the norms of $\dot{\mathbf{b}}_x$, $\boldsymbol{\omega}_{L_x}$, and $\boldsymbol{\omega}_{U_x}$, respectively:

$$\|\dot{\mathbf{b}}_x\| = \dot{x}_{EE} \quad (4-34)$$

$$\|\boldsymbol{\omega}_{L_x}\| = \frac{(\dot{y}_{EE}\mathbf{e}_2 + \dot{z}_{EE}\mathbf{e}_3) \cdot \mathbf{U}_x}{(\mathbf{e}_1 \times \mathbf{L}_x) \cdot \mathbf{U}_x} \quad (4-35)$$

$$\|\boldsymbol{\omega}_{U_x}\| = \frac{(\dot{y}_{EE}\mathbf{e}_2 + \dot{z}_{EE}\mathbf{e}_3) \cdot \mathbf{L}_x}{(\mathbf{e}_1 \times \mathbf{U}_x) \cdot \mathbf{L}_x} \quad (4-36)$$

Similarly, for the y and z legs, the norms of $\dot{\mathbf{b}}_y$, $\boldsymbol{\omega}_{L_y}$, $\boldsymbol{\omega}_{U_y}$, $\dot{\mathbf{b}}_z$, $\boldsymbol{\omega}_{L_z}$, and $\boldsymbol{\omega}_{U_z}$ can be obtained by changing the subscript to the corresponding one since the three legs are identical.

Acceleration analysis

The reverse acceleration analysis can use similar steps which are in velocity section. As for the x leg, the acceleration of the EE and the derivative of Equations 4-31 and 4-32 with respect to time by using triple product expansion can be expressed respectively as [12]:

$$\mathbf{a}_{EE} = (\ddot{x}_{EE}, \ddot{y}_{EE}, \ddot{z}_{EE})^T = \ddot{x}_{EE}\mathbf{e}_1 + \ddot{y}_{EE}\mathbf{e}_2 + \ddot{z}_{EE}\mathbf{e}_3 \quad (4-37)$$

$$\ddot{\mathbf{L}}_x = \dot{\boldsymbol{\omega}}_{L_x} \times \mathbf{L}_x - \|\boldsymbol{\omega}_{L_x}\|^2 \mathbf{L}_x \quad (4-38)$$

$$\ddot{\mathbf{U}}_x = \dot{\boldsymbol{\omega}}_{U_x} \times \mathbf{U}_x - \|\boldsymbol{\omega}_{U_x}\|^2 \mathbf{U}_x \quad (4-39)$$

Taking the time derivative of Equation 4-33 and substituting the corresponding terms by Equations 4-37, 4-38 and 4-39 will give:

$$\begin{aligned} & \ddot{\mathbf{b}}_x + \dot{\boldsymbol{\omega}}_{L_x} \times \mathbf{L}_x - \|\boldsymbol{\omega}_{L_x}\|^2 \mathbf{L}_x + \dot{\boldsymbol{\omega}}_{U_x} \times \mathbf{U}_x \\ & - \|\boldsymbol{\omega}_{U_x}\|^2 \mathbf{U}_x - (\ddot{x}_{EE}\mathbf{e}_1 + \ddot{y}_{EE}\mathbf{e}_2 + \ddot{z}_{EE}\mathbf{e}_3) = \mathbf{0} \end{aligned} \quad (4-40)$$

Then taking the dot product of $\mathbf{e}_1$, $\mathbf{U}_x$, and $\mathbf{L}_x$ on both sides of Equation 4-40 and using the property of the scalar triple product will respectively get the norms of linear acceleration vector of the prismatic joint and of the angular acceleration vectors of the lower and upper links:

$$\|\ddot{\mathbf{b}}_x\| = \ddot{x}_{EE} \quad (4-41)$$

$$\|\dot{\boldsymbol{\omega}}_{L_x}\| = \frac{\left((\ddot{y}_{EE}\mathbf{e}_2 + \ddot{z}_{EE}\mathbf{e}_3) + \|\boldsymbol{\omega}_{L_x}\|^2 \mathbf{L}_x + \|\boldsymbol{\omega}_{U_x}\|^2 \mathbf{U}_x \right) \cdot \mathbf{U}_x}{(\mathbf{e}_1 \times \mathbf{L}_x) \cdot \mathbf{U}_x} \quad (4-42)$$

$$\|\dot{\omega}_{U_x}\| = \frac{\left((\ddot{y}_{EE} \mathbf{e}_2 + \ddot{z}_{EE} \mathbf{e}_3) + \|\omega_{L_x}\|^2 \mathbf{L}_x + \|\omega_{U_x}\|^2 \mathbf{U}_x \right) \cdot \mathbf{L}_x}{(\mathbf{e}_1 \times \mathbf{U}_x) \cdot \mathbf{L}_x} \quad (4-43)$$

The whole acceleration analysis approach of the x leg is also applicable for the y and z legs.

Reverse Force Analysis

As designing a prototype of the Tripteron, the three linear actuators' specifications should be known. The most important parameter is the force that each actuator can apply. When an external wrench is applied to the EE, three linear actuator forces can be obtained. Furthermore, there are reaction forces and moments at the revolute joints due to the external wrench.

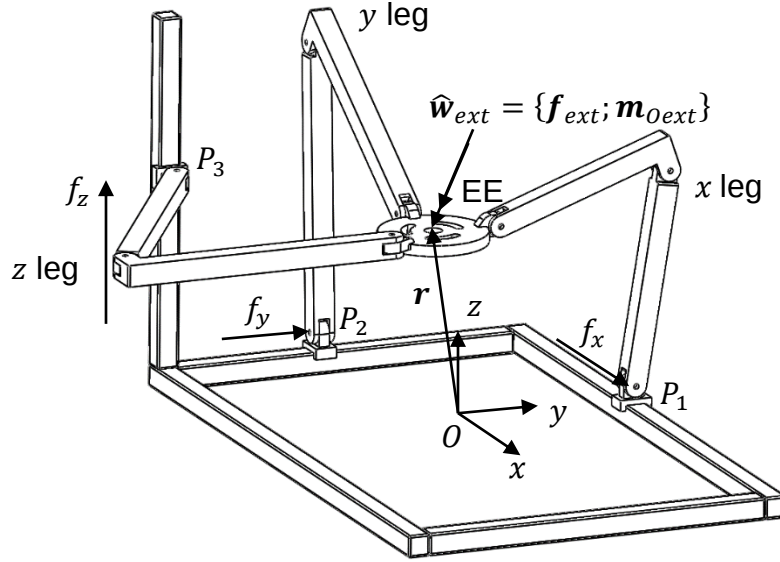


Figure 4-4. An external wrench applied to the end-effector of the Tripteron.

As shown in Figure 4-4, an external wrench $\hat{\mathbf{w}}_{ext} = \{\mathbf{f}_{ext}; \mathbf{m}_{oext}\}$ [11] applied to the EE and the position vector $\mathbf{r}$ of the EE is given. Gravity effects on connecting links are ignored to simplify the analysis.

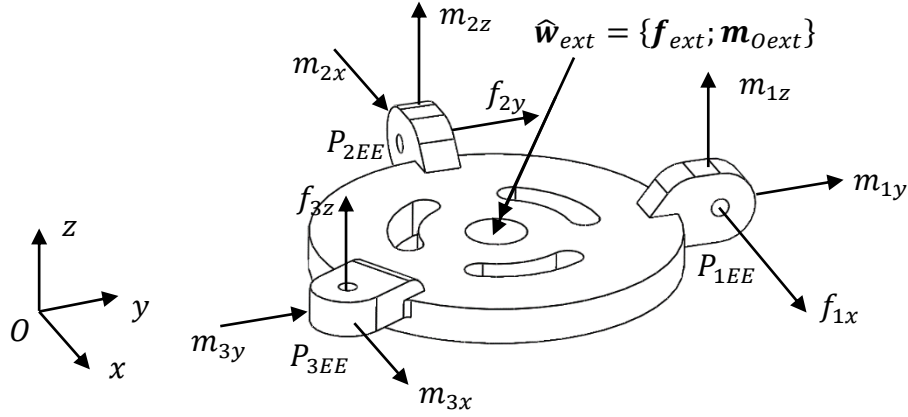


Figure 4-5. Free body diagram of the end-effector.

To get all the unknown components, the free body diagram of the EE is shown in Figure 4-5. Since constant mechanism parameters are known and the position vector $\mathbf{r}$ is given, the coordinates of points P_{1EE} , P_{2EE} , and P_{3EE} can be calculated and written as:

$$P_{iEE} = (x_{iEE}, y_{iEE}, z_{iEE})^T, i = 1, 2, 3 \quad (4-44)$$

According to the free body diagram, the x leg has a force passing through point P_{1EE} in the x direction. There is no force component in the y or z directions as the three revolute joints of the x leg cannot provide any torque that would be necessary to balance these force components. As for the moment, there is no moment component along the x direction that the x leg can be applying to the EE since the three revolute joints of the x leg are passive. Therefore, the wrench acting on the EE as applied by the x leg can be written as:

$$\begin{Bmatrix} \mathbf{f}_1 \\ \mathbf{m}_{01} \end{Bmatrix} = \begin{Bmatrix} f_{1x} \\ 0 \\ 0 \\ 0 \\ m_{1y} + f_{1x}z_{1EE} \\ m_{1z} - f_{1x}y_{1EE} \end{Bmatrix} \quad (4-45)$$

Since similar arguments can be made for the y and z legs, the wrenches can be respectively expressed as:

$$\begin{Bmatrix} \mathbf{f}_2 \\ \mathbf{m}_{02} \end{Bmatrix} = \begin{Bmatrix} 0 \\ f_{2y} \\ 0 \\ m_{2x} - f_{2y}z_{2EE} \\ 0 \\ m_{2z} + f_{2y}x_{2EE} \end{Bmatrix} \quad (4-46)$$

$$\begin{Bmatrix} \mathbf{f}_3 \\ \mathbf{m}_{03} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ f_{3z} \\ m_{3x} + f_{3z}y_{3EE} \\ m_{3y} - f_{3z}x_{3EE} \\ 0 \end{Bmatrix} \quad (4-47)$$

Equating the leg wrenches to the external wrench gives:

$$\begin{Bmatrix} f_{ext_x} \\ f_{ext_y} \\ f_{ext_z} \\ m_{0ext_x} \\ m_{0ext_y} \\ m_{0ext_z} \end{Bmatrix} = \begin{Bmatrix} f_{1x} \\ f_{2y} \\ f_{3z} \\ m_{2x} + m_{3x} + f_{3z}y_{3EE} - f_{2y}z_{2EE} \\ m_{1y} + m_{3y} + f_{1x}z_{1EE} - f_{3z}x_{3EE} \\ m_{1z} + m_{2z} + f_{2y}x_{2EE} - f_{1x}y_{1EE} \end{Bmatrix} \quad (4-48)$$

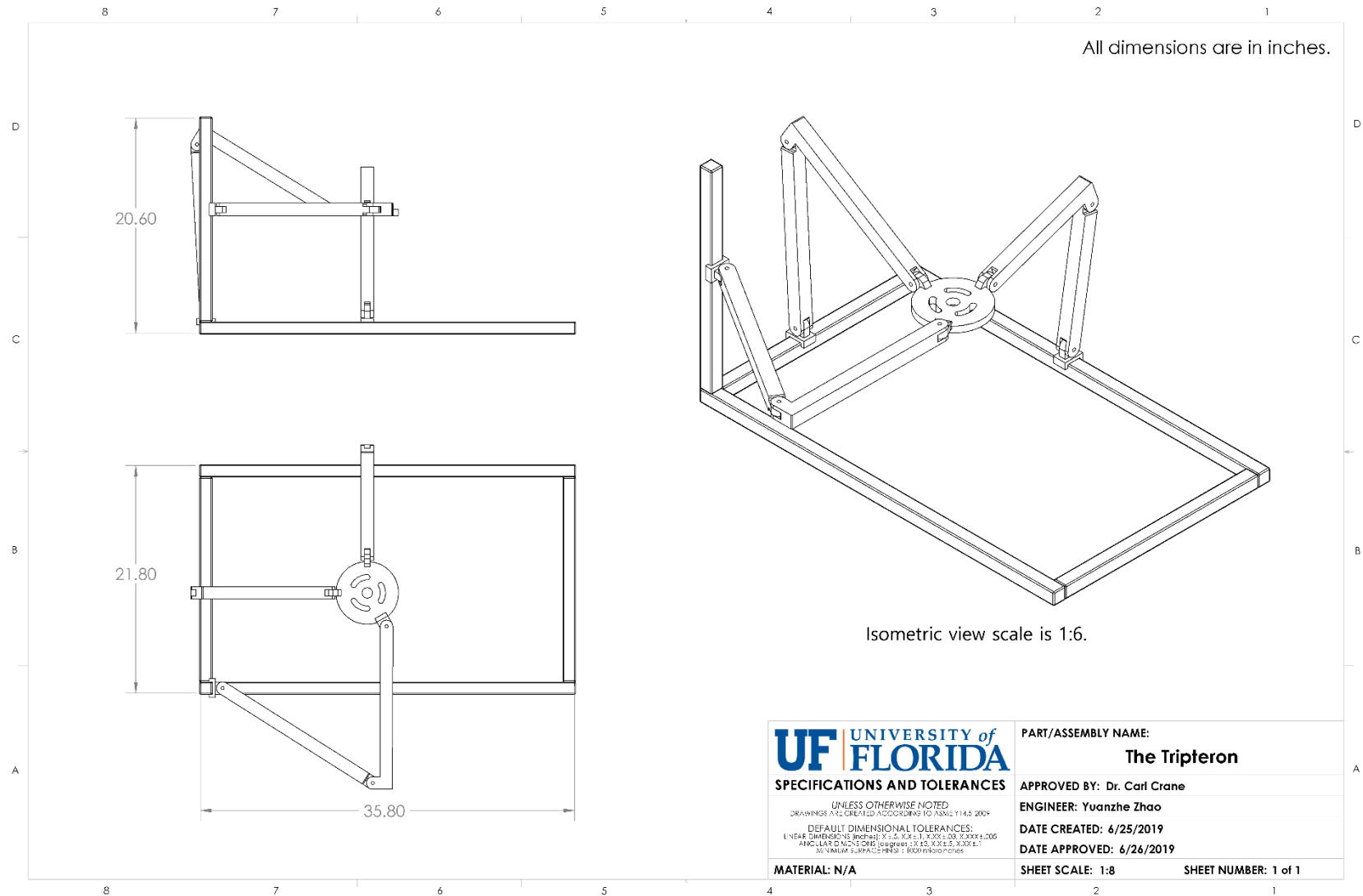
Then, the three force values f_{1x} , f_{2y} , and f_{3z} are determined from the top three components. The moment equations, however, give three equations in the six unknowns m_{1y} , m_{1z} , m_{2x} , m_{2z} , m_{3x} , and m_{3y} . Since no other equations can be obtained, the six unknown moments at the EE are statically indeterminate. Therefore, the magnitudes of three linear actuator forces $\mathbf{f}_x$, $\mathbf{f}_y$, and $\mathbf{f}_z$ are equal to the three force values f_{1x} , f_{2y} , and f_{3z} .

CHAPTER 5 CONCLUSION

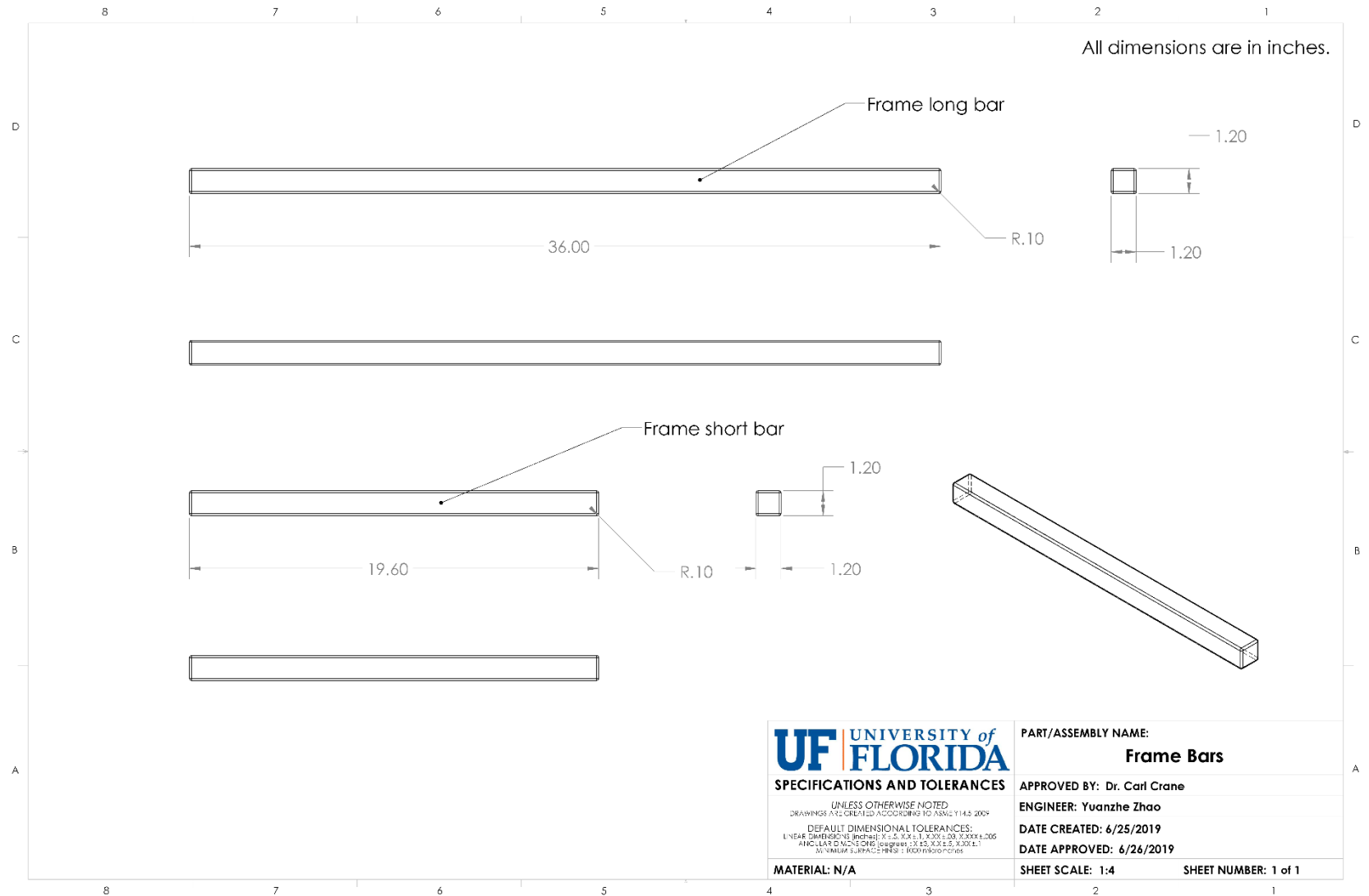
This thesis presented a design and the 3D model of a 3-DOF parallel translational mechanism and discussed the position and force analysis of this device. The positions of the mechanism are notated by vectors and a closed-loop vector equation can be established by the vectors of each part. Therefore, the reverse displacement, velocity and acceleration analysis was solved by the closed-loop equation and geometric relationships. For the reverse force analysis, the three linear actuator forces were determined from the three components of the external force applied to the end-effector. Furthermore, since the six unknown moments cannot be solved when an external wrench applied to the end-effector, the mechanism is statically indeterminate for the reaction moments.

APPENDIX
DESIGN DRAWING

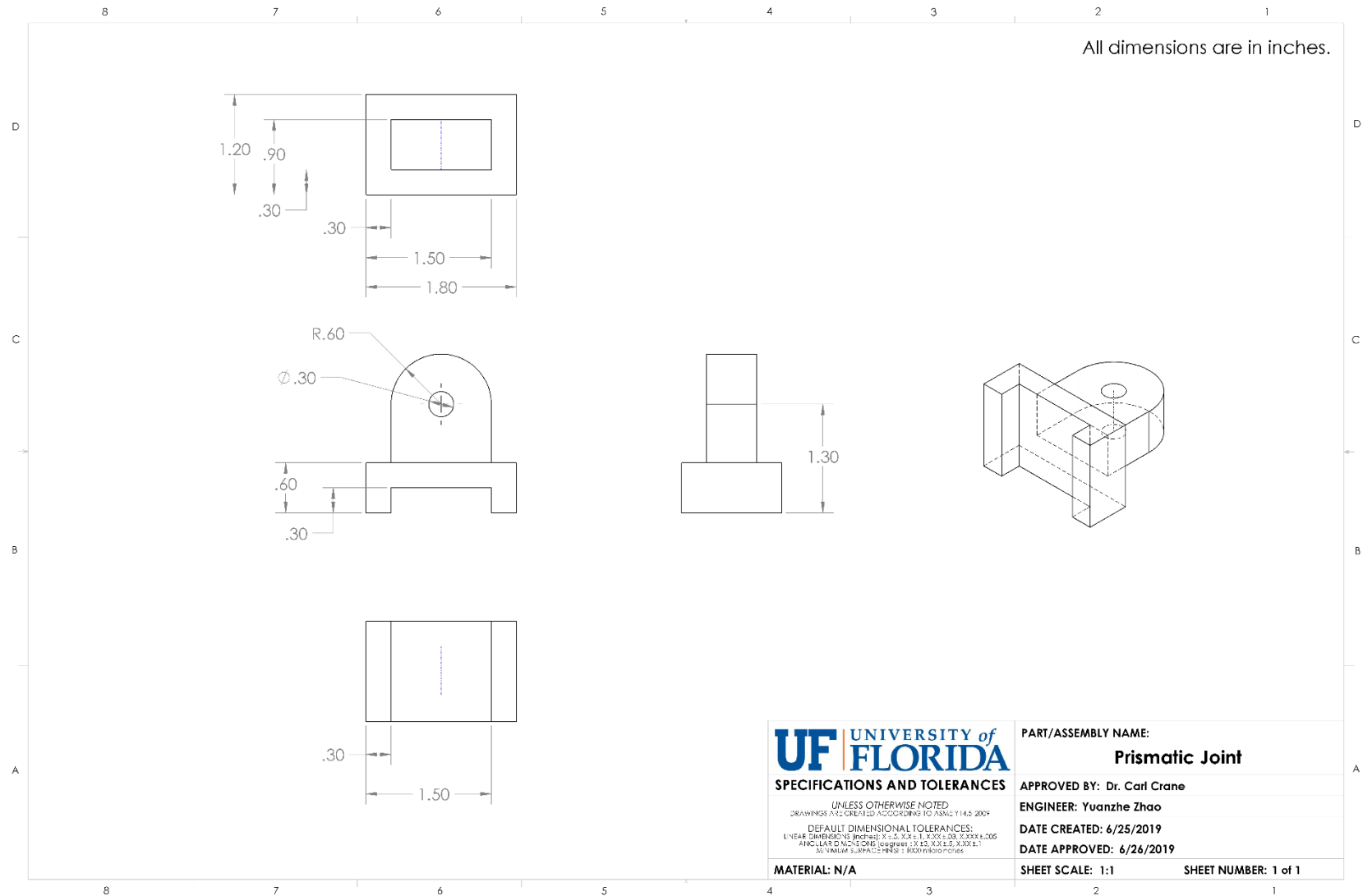
1. Frame



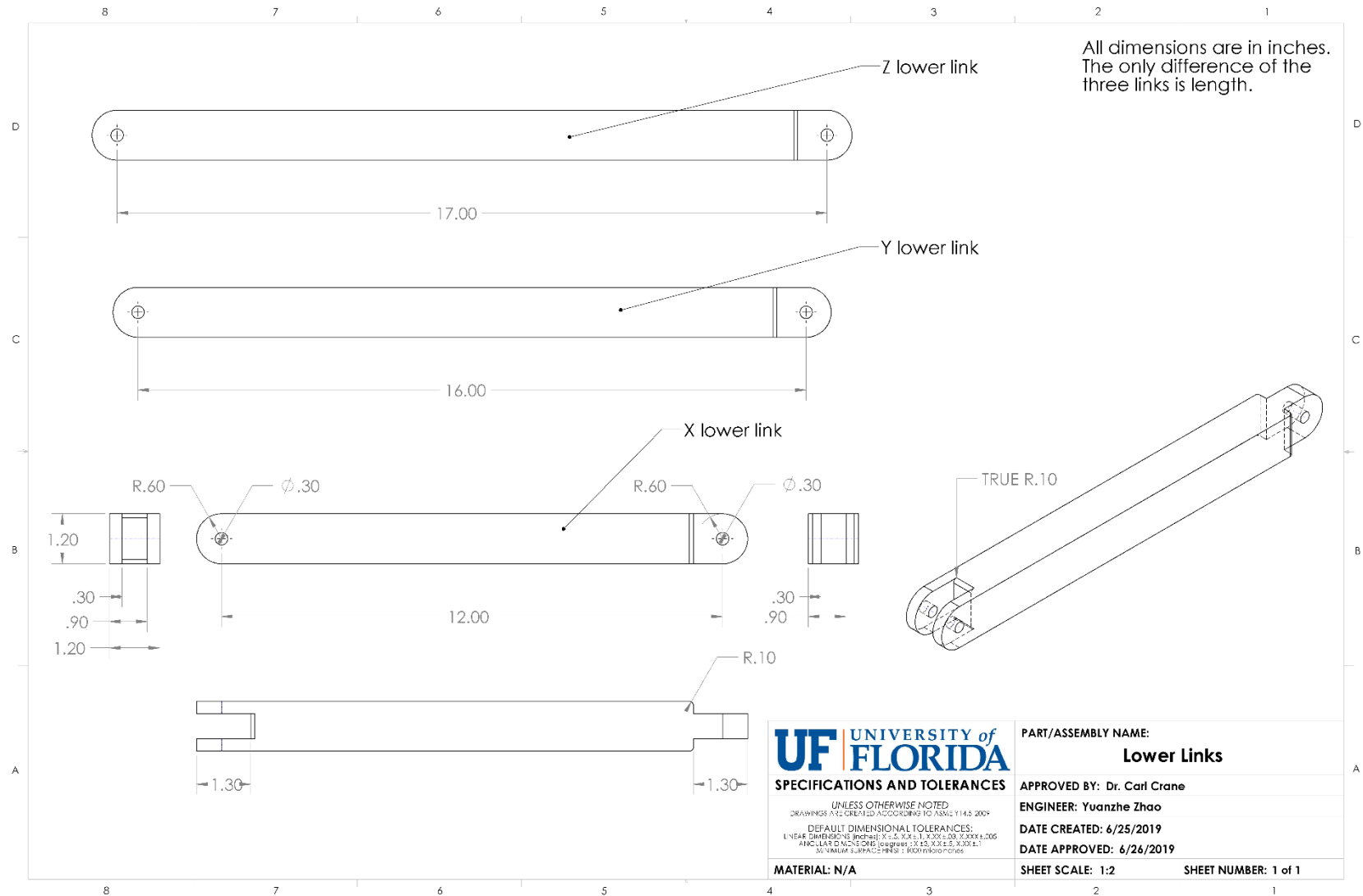
2. Frame Bars



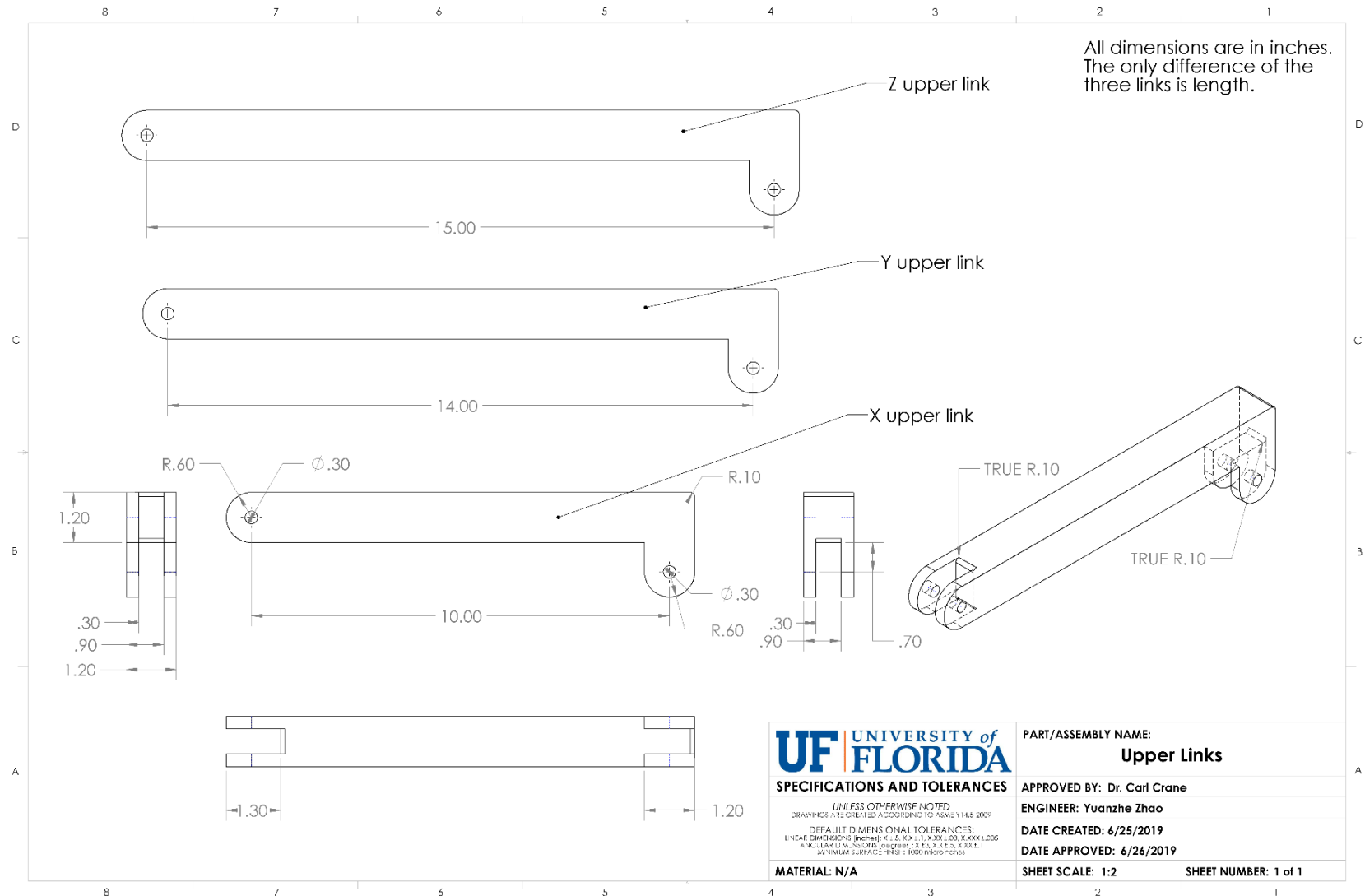
3. Prismatic Joint



4. Lower Links



5. Upper Links



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BIOGRAPHICAL SKETCH

Yuanzhe Zhao was born in Luoyang, Henan, China in 1995. He completed his Bachelor of Science degree in energy and power engineering in the School of Mechanical Engineering at Shanghai Jiao Tong University, Shanghai, China in June 2017. He was also awarded the first prize of National Mathematical Olympiad (Provincial-Level) during high school and Academic Excellence Scholarship (Third-Class) of SJTU for 2015-2016 academic year. Though he majored in energy and power engineering, he was more interested in robotics and control systems. With this enthusiasm, he joined the Department of Mechanical and Aerospace Engineering at University of Florida for a Master of Science degree in mechanical engineering and focusing on Dynamics, Controls, and Systems. During his graduate education, he has been working on his master's thesis with Dr. Carl Crane for analyzing the Tripteron, a 3-DOF parallel translational mechanism.